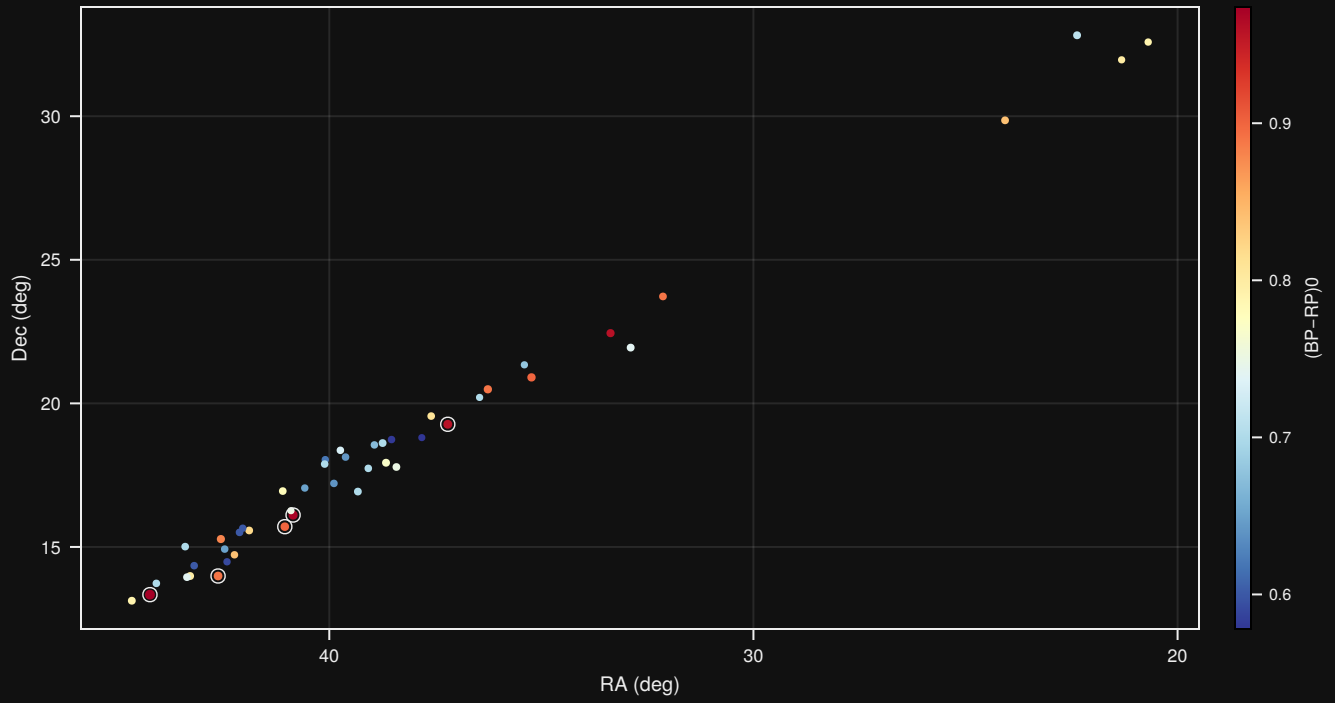
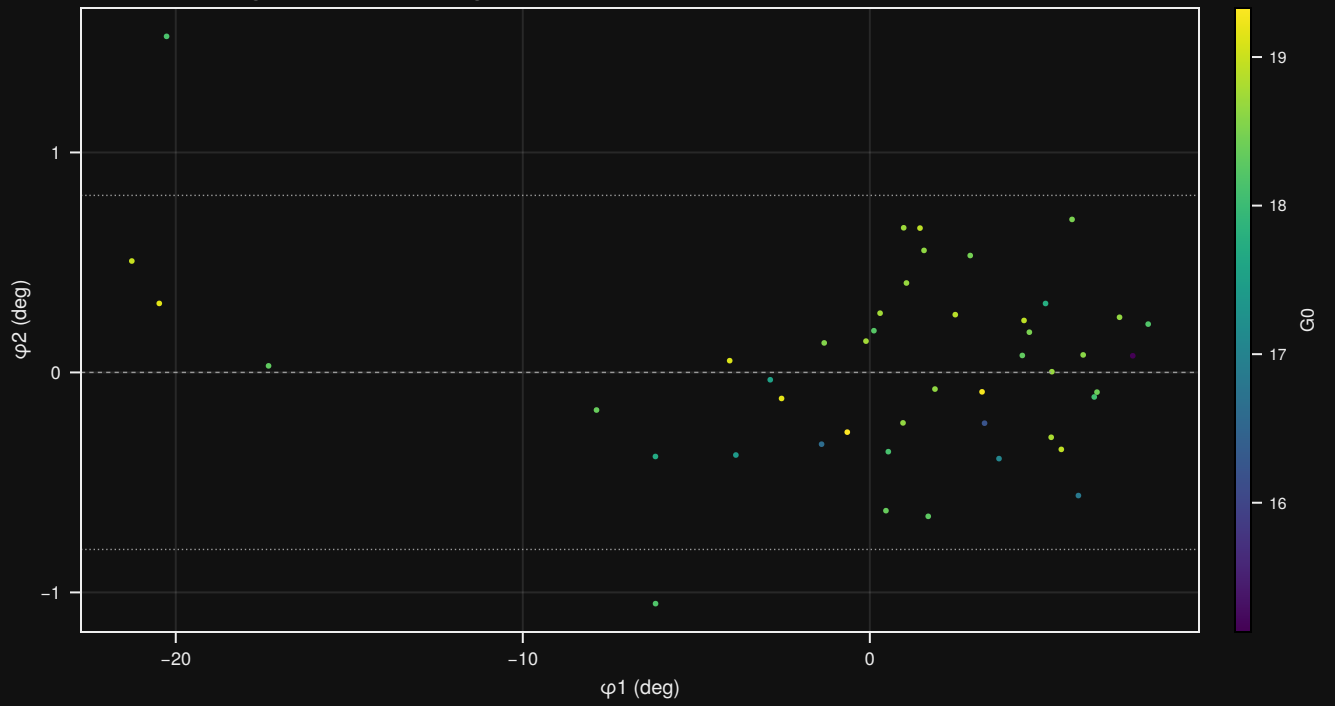
Gaia-12

MEMBERSHIP	
Stars in catalog	46
with Gaia GSP-Phot	34 (74%)
with radial velocity	5
Stream id (sID)	10
GEOMETRY · great-circle fit	
Centroid RA, Dec	38.3, 17.8 deg
Centroid l, b	154.6, -38.4 deg
Pole RA, Dec	147.2, 43.6 deg
Length (φ1, 2–98%)	29.3 deg
Width σ(φ2), robust	0.40 deg
KINEMATICS	
pmRA* med ± σ	5.56 ± 0.64 mas/yr
pmDE med ± σ	-5.46 ± 0.60 mas/yr
μφ1* med ± σ	7.71 ± 0.85 mas/yr
μφ2 med ± σ	-0.13 ± 0.30 mas/yr
Parallax (cat.) med	0.138 ± 0.282 mas
Parallax (wtd, ZP-corr.)	0.096 ± 0.026 mas
Distance	d = 10.4 ± 2.8 kpc
Physical length	5.30 kpc
RV median ± σ(MAD)	-280.8 ± 24.3 km/s
RV gradient dV/dφ1	6.36 km/s/deg
PHOTOMETRY · POPULATIONS	
G0 range (median)	15.1–19.3 (18.5)
(BP−RP)0 median	0.74
GSP-Phot [M/H] (n=34)	-2.37 ± 0.78
GSP-Phot Teff median	5468 K
RUWE > 1.4	0.0 %
PM clusters (largest)	3 (14 stars)
IBATA+2024 · TABLE 3	
n catalog | paper	46 | 46
n_RV catalog | paper	5 | 5
<[Fe/H]> paper	-3.28
Stellar mass (paper)	1200 Msun
Pole offset vs paper	0.3 deg

Sky map · catalog members (n = 46) · size ∝ brightness · ring = has RV

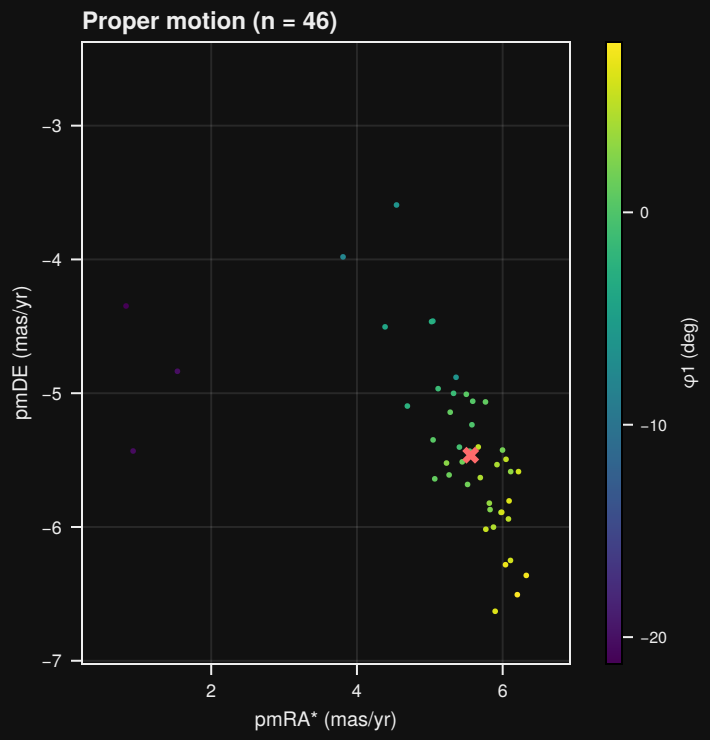


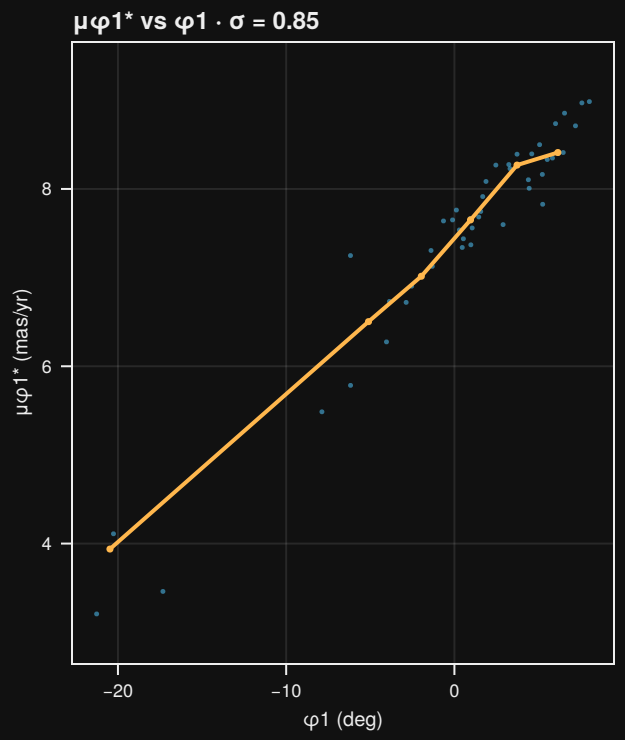
Stream frame (great-circle fit) · length 29.3° · width σ 0.40°

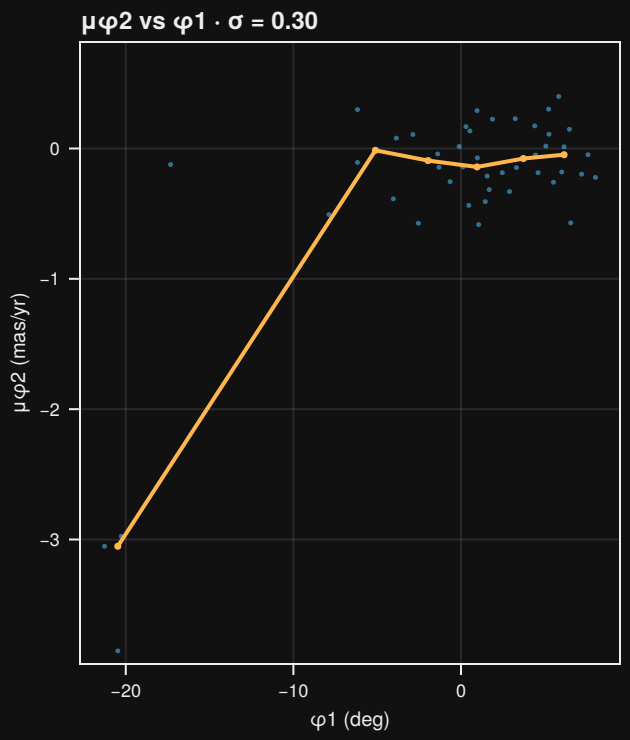


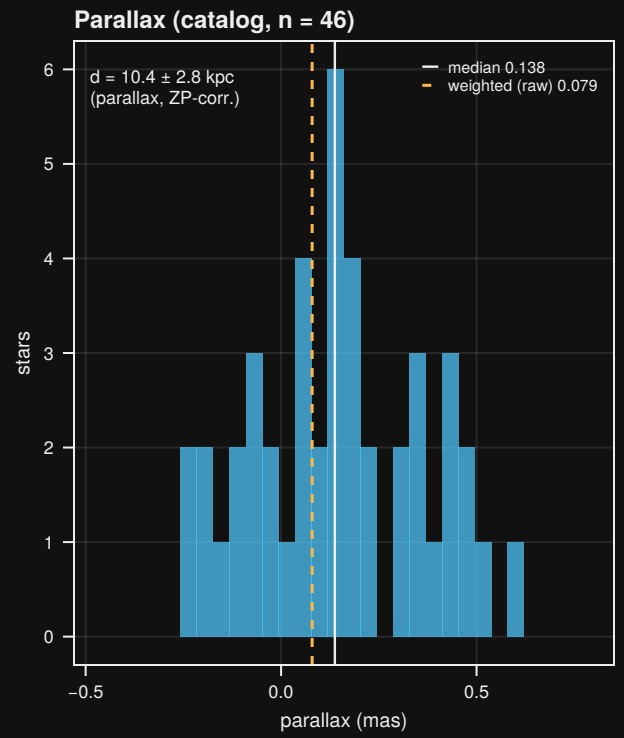
Gaia-12

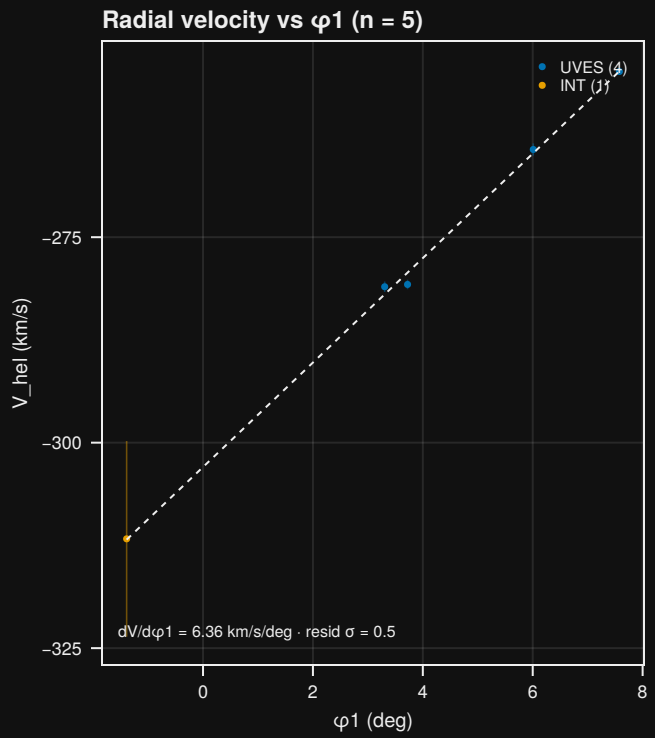
Kinematics · proper motion · parallax · radial velocity

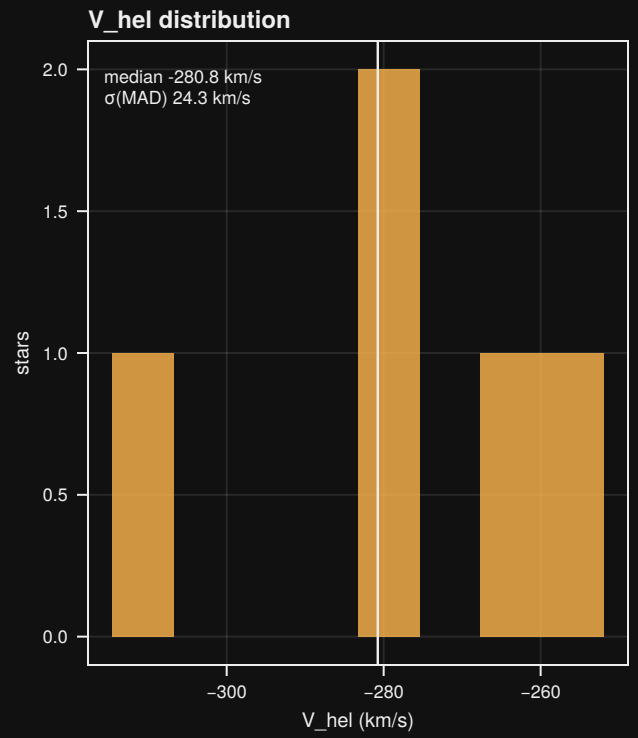








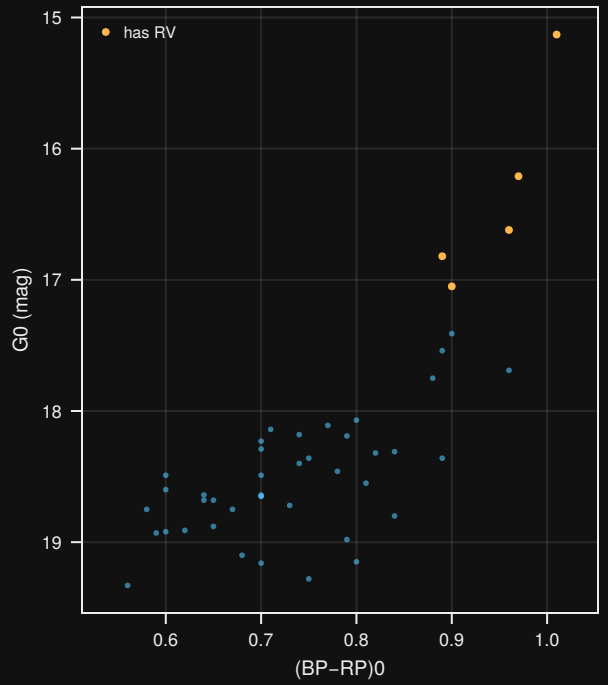




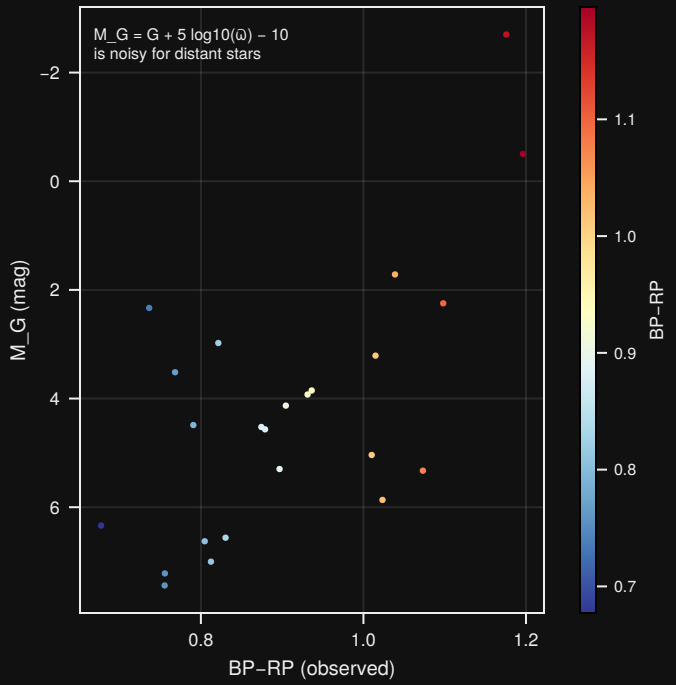
Gaia-12

Photometry · stellar types

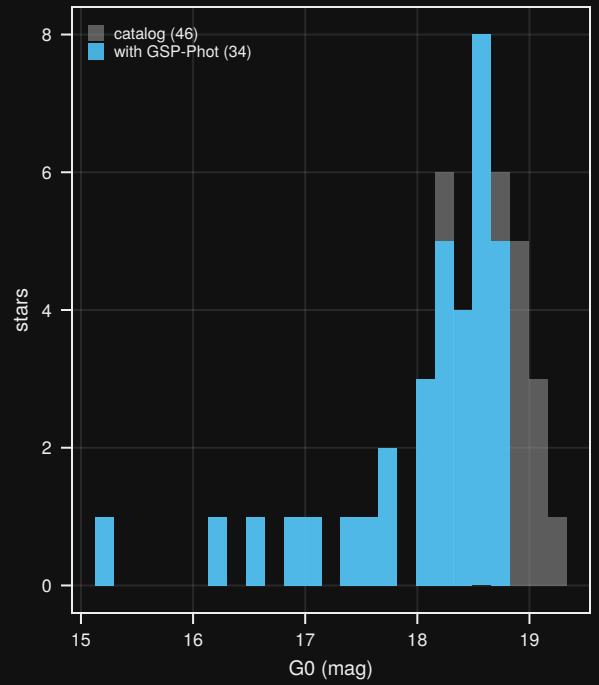
CMD · dereddened (n = 46)



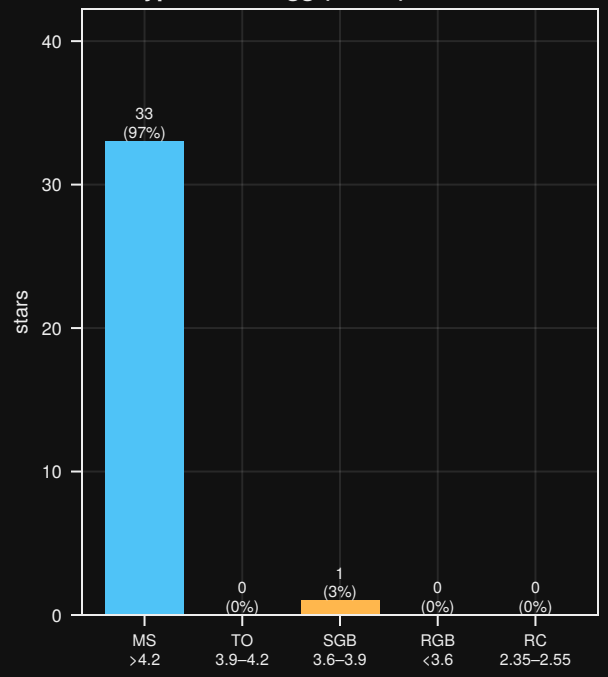
HR diagram · ϖ>0 (n = 24/34)



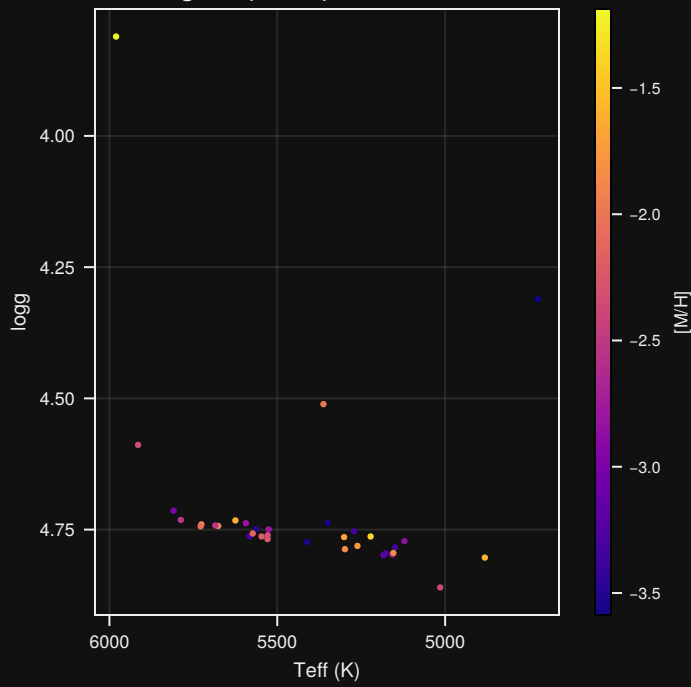
G0 & GSP-Phot coverage



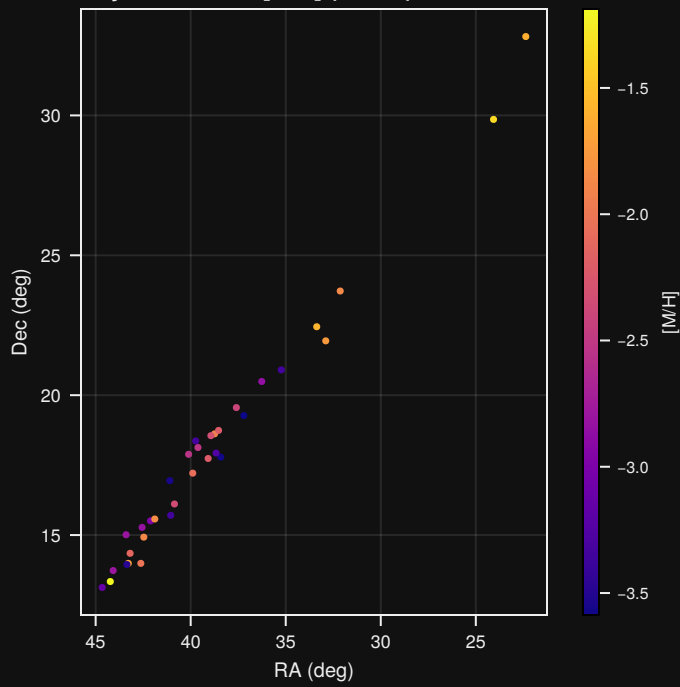
Star types from logg (n = 34)



Kiel diagram (n = 34)



Sky · GSP-Phot [M/H] (n = 34)

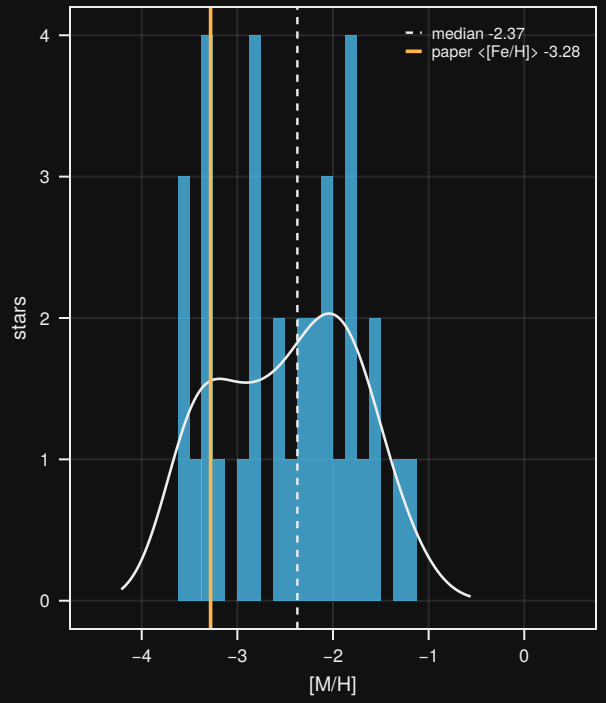


GSP-Phot (Gaia low-resolution BP/RP) parameters are prior-dominated for faint, distant halo stars (G ≥ 17): logg, [M/H] and ages are indicative only and typically biased toward dwarfs.

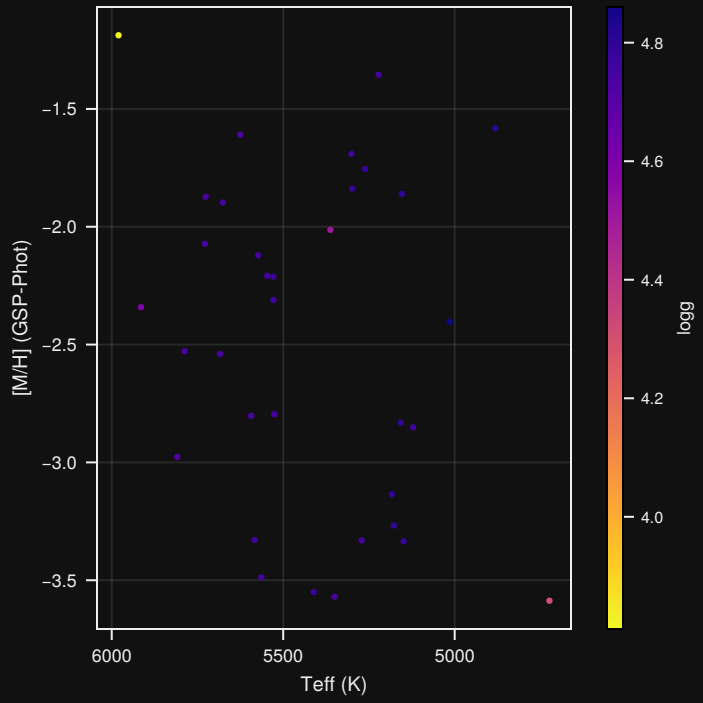
Gaia-12

Metallicity · age · astrometric quality

[M/H] GSP-Phot (n = 34)



Teff vs [M/H] (n = 34)



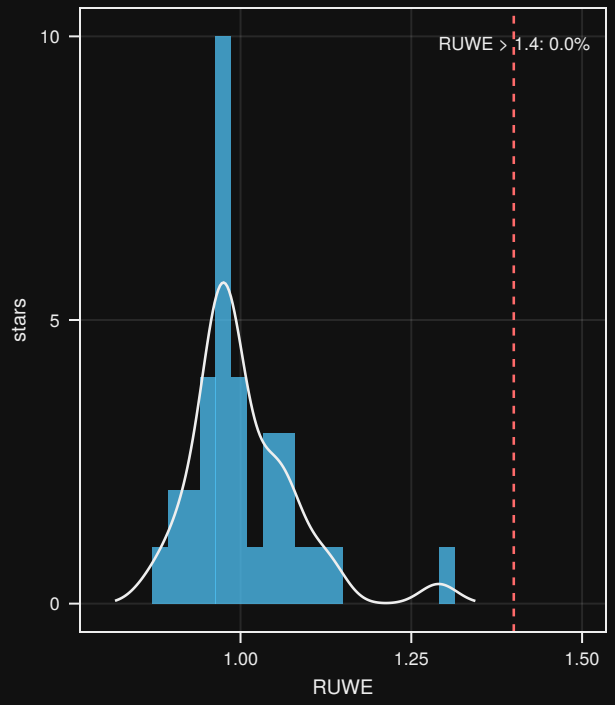
FLAME age (n = 2)

FLAME ages for only 2 star(s)
(FLAME needs good parallaxes)

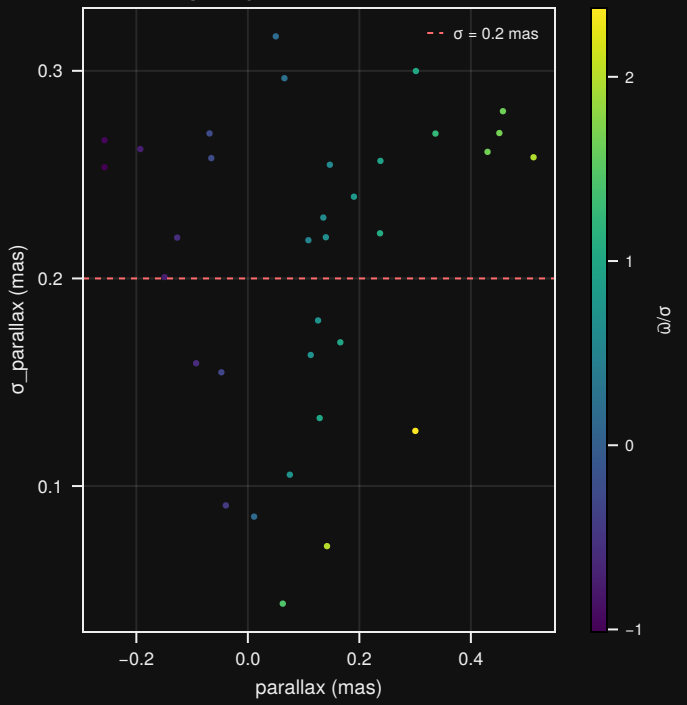
Age–metallicity (n = 2)

n < 3

RUWE (n = 34)

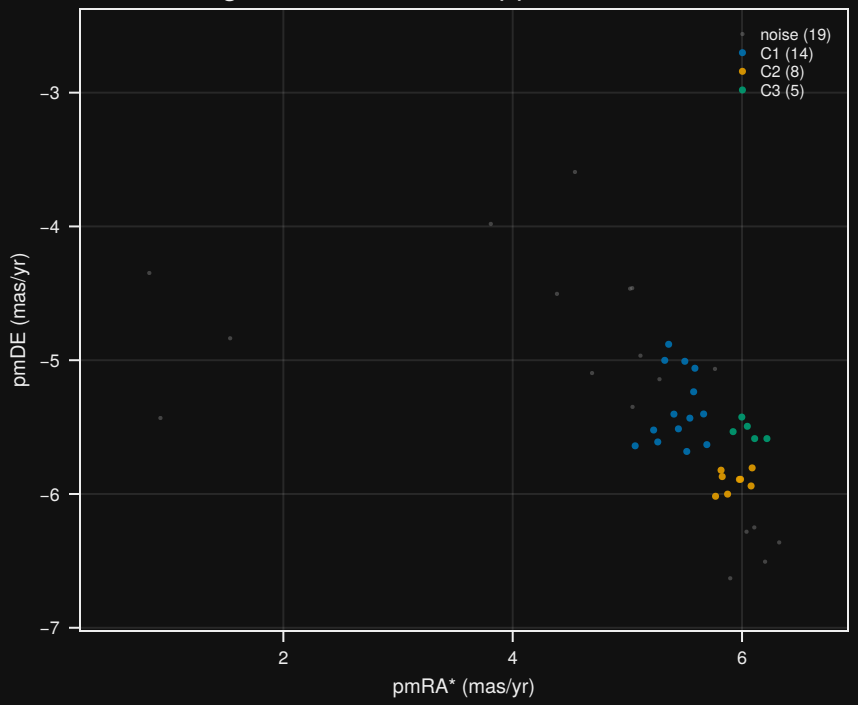


Parallax quality (n = 34)

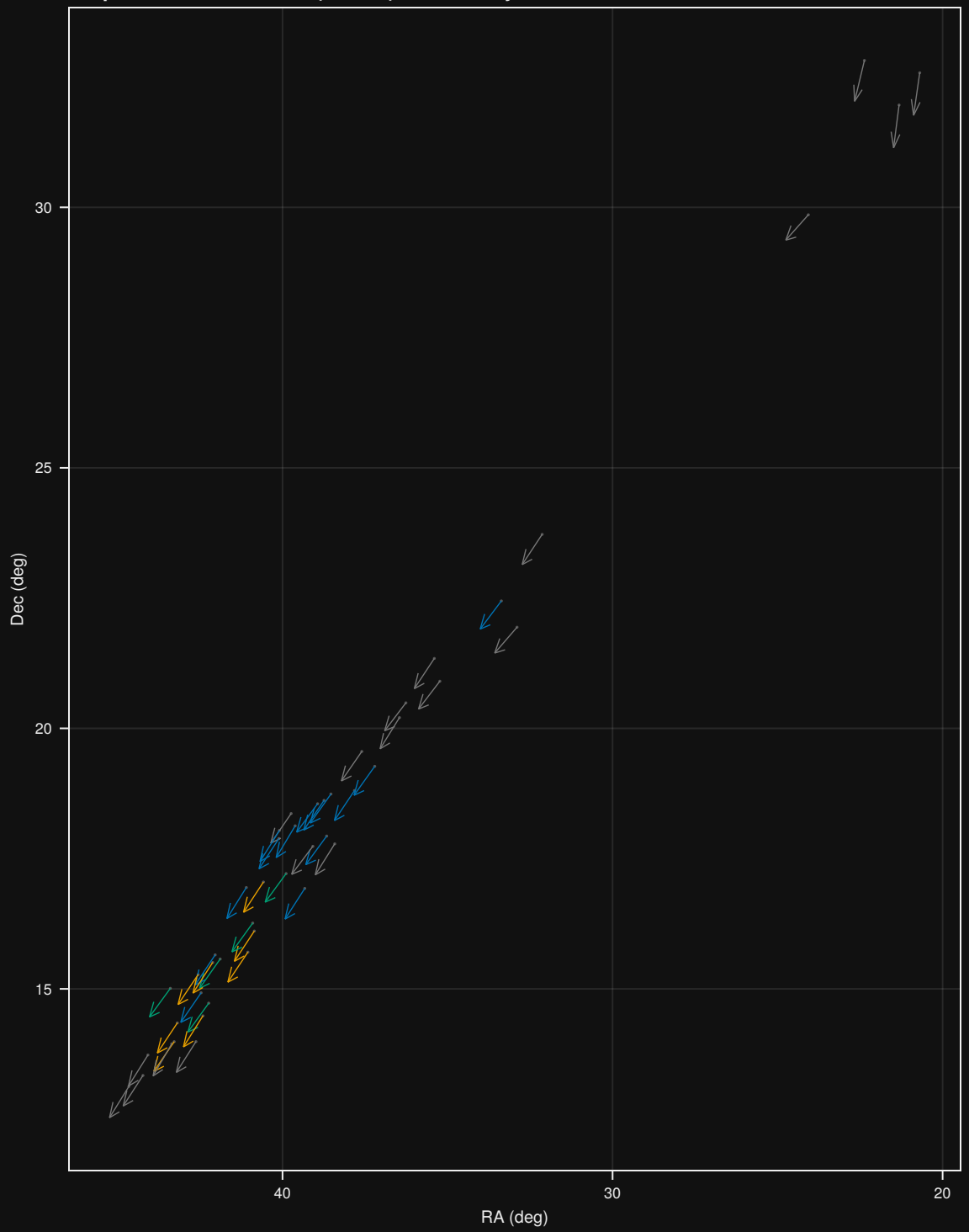


Gaia-12

PM clustering · HDBSCAN · 3 cluster(s) · noise 41%



Proper-motion directions (arrows) coloured by cluster



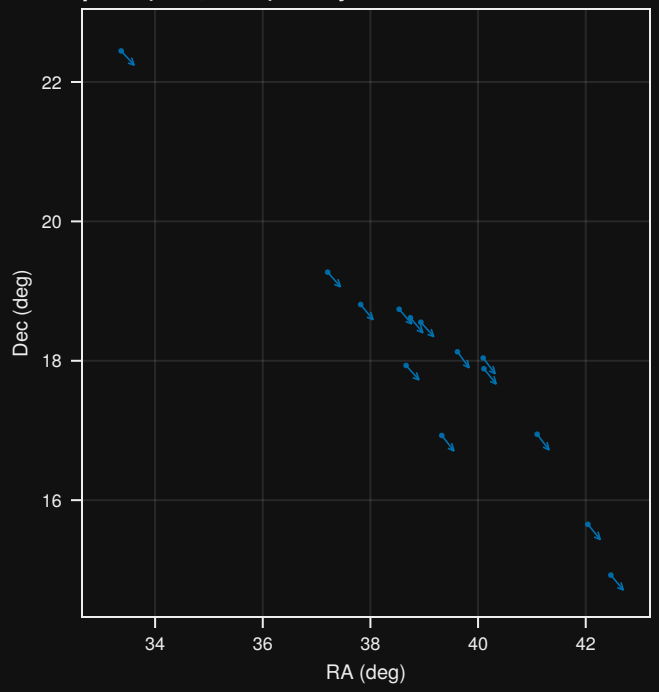
Cluster summary

id	n	pmRA	pmDE	σpmRA	[M/H]	n_RV
noise	19	5.04	-5.07	1.27	-2.40	2
C1	14	5.47	-5.42	0.17	-2.42	1
C2	8	5.93	-5.89	0.15	-2.59	2
C3	5	6.05	-5.53	0.09	-2.07	0

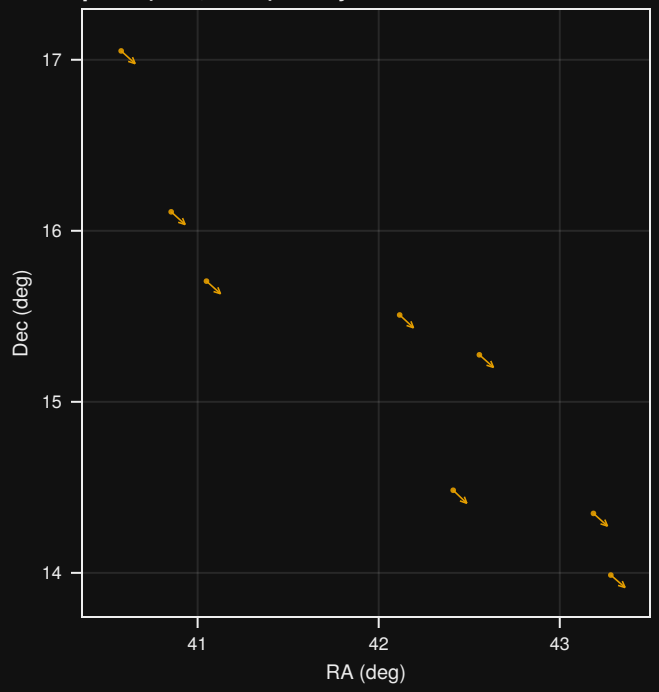
Gaia-12

Per-cluster proper-motion vector maps

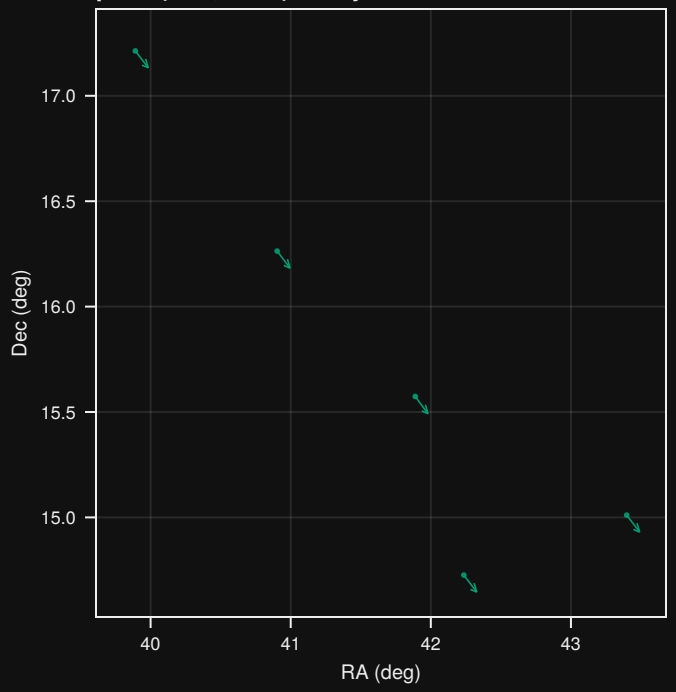
Cluster C1 · n = 14
pm = (5.47, -5.42) mas/yr



Cluster C2 · n = 8
pm = (5.93, -5.89) mas/yr



Cluster C3 · n = 5
pm = (6.05, -5.53) mas/yr



Noise · n = 19

